

Unifying Special Relativity and Quantum Mechanics

Lecture 10 — Completing the Massive Vector Field (Propagator, Feynman Rule, Helmholtz Decomposition) and the Massless Case: the Photon Field

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–9: everything in the normal text and in the **yellow / pink** boxes is a faithful transcription of the handwritten notes (the coloured boxes reproduce the results that were framed or highlighted in the original — yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones). Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, and connections to the previous lectures.

Pagination of this scan. This scan contains *two* blocks of pages.

- (a) Sheets carrying the circled numbers 18–25. These continue the chapter that Lecture 9 covered (whose pages were marked [p. 3****]–[p. 18****]), so they are written here as [p. 18****]–[p. 25****].
- (b) Sheets carrying the circled numbers 1–9: the notes open a new chapter and restart the page numbering from 1 for the *fifth* time. To keep these apart from all the earlier restarts they are marked [p. 1*****]–[p. 9*****] (five stars).

Overlap with Lecture 9. The first sheet, [p. 18****], is a *repeat*: it is the very last page of the Lecture 9 scan. It is reproduced here in full (Section 1) because the whole of Lecture 10 starts from the question it leaves open.

Where we left off (recap of Lecture 9). For the massive spin-1 field

$$V^\mu(x) = \sum_{\sigma} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\kappa u^\mu(\vec{p}, \sigma) e^{-ip \cdot x} a(\vec{p}, \sigma) + \eta v^\mu(\vec{p}, \sigma) e^{ip \cdot x} a^\dagger(\vec{p}, \sigma) \right],$$

the polarization vectors were constructed from the little group $SO(3)$, the polarization sum was found to be $\sum_{\lambda} \varepsilon^\mu \varepsilon^{\nu*} = -g^{\mu\nu} + p^\mu p^\nu / M^2$, and the equal-time bracket $[V^\mu(x), V^\nu(y)]_{\pm}$ was reduced to $(-g^{\mu\nu} - \frac{1}{M^2} \partial_{x_\mu}^2 \partial_{x_\nu}^2) \kappa \eta [\Delta_+(x-y) \pm \Delta_+(y-x)]$. This lecture completes the story (statistics, κ , η , the propagator, the Feynman rule, the counting of degrees of freedom) and then restarts the whole programme for *massless* particles.

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Part I — Completing the massive spin-1 field

1 Microcausality of $[V^\mu, V^\nu]$: statistics is fixed [p. 18****]

The function $\Delta_+(x - y)$ enjoys the property that

$$\Delta_+(x - y) = \Delta_+(y - x) \quad \text{if } (x - y)^2 < 0.$$

Consequently, if $(x - y)^2 < 0$, we write

$$[V^\mu(x), V^\nu(y)]_\pm = \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \kappa \eta \underbrace{\left[\Delta_+(x - y) \pm \Delta_+(y - x) \right]}_{\substack{\text{the only possibility to get zero} \\ \text{is to pick the minus sign}}} \stackrel{!}{=} 0.$$

► Annihilation and creation operators must obey commutation relations in order to have a field theory description of massive spin-1 particles compatible with relativistic causality.

► At this stage, κ and η are still arbitrary complex numbers. However, we still have to impose the microcausality condition

$$[V^\mu(x), V^\nu(y)^\dagger] = 0 \quad \text{if } (x - y)^2 < 0.$$

Added explanation (not in the original notes)

Two transcription remarks. (i) As on [p. 17****] of Lecture 9, the factor $1/M^2$ in front of the double derivative is dropped in the handwriting; it has been restored above, since $p^\mu p^\nu \rightarrow -\frac{1}{M^2} \partial^2 / \partial x_\mu \partial x_\nu$ came from the polarization sum $-g^{\mu\nu} + p^\mu p^\nu / M^2$. It plays no role in the argument: the whole differential operator multiplies the bracket and cannot change whether the bracket vanishes. (ii) The bracket in the second boxed equation is now a *commutator* (no $\pm$), because the first line of this page has just decided the statistics.

This is the spin–statistics theorem in the spin-1 case. Nothing about statistics was assumed: causality alone selected the commutator, i.e. *Bose* statistics. The reason is that the spin-1 polarization sum is *even* under $p \rightarrow -p$, so the two Wightman functions $\Delta_+(x - y)$ and $\Delta_+(y - x)$ enter with the same sign; outside the light cone they are equal, so only the minus sign in $[\cdot, \cdot]_-$ can kill them. For spin 1/2 the analogous spin sum $\sum_s u \bar{u} = p_\mu \gamma^\mu + m$ is *odd* in p^μ , the relative sign flips, and causality then demands the *anticommutator*. Integer spin $\rightarrow$ Bose, half-integer spin $\rightarrow$ Fermi.

What is still missing. Microcausality has two halves (Lecture 7, [p. 26***]): $[V, V]$ and $[V, V^\dagger]$. The first involved the *product* $\kappa \eta$ and therefore said nothing about the magnitudes. The second, worked out on the next page, involves $|\kappa|^2$ and $|\eta|^2$ separately — and that is what will force $|\kappa| = |\eta|$, the statement that a particle and its antiparticle must be created/destroyed with exactly equal weight.

2 The second microcausality condition: $\kappa = \eta$ [p. 19****]

$$[V^\mu(x), V^\nu(y)^\dagger] = \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right) \left[|\kappa|^2 e^{-ip \cdot (x-y)} - |\eta|^2 e^{ip \cdot (x-y)} \right]$$

$$\begin{aligned}
&= \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} \left[|\kappa|^2 e^{-ip \cdot (x-y)} - |\eta|^2 e^{ip \cdot (x-y)} \right] \\
&= \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \left[|\kappa|^2 \Delta_+(x-y) - |\eta|^2 \Delta_+(y-x) \right].
\end{aligned}$$

If we now require $(x-y)^2 < 0$, we have $\Delta_+(x-y) = \Delta_+(y-x)$:

$$[V^\mu(x), V^\nu(y)^\dagger] = \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \underbrace{(|\kappa|^2 - |\eta|^2)}_{\rightsquigarrow |\kappa|^2 = |\eta|^2} \Delta_+(x-y) \stackrel{!}{=} 0$$

$$|\kappa| = |\eta|$$

► If we exploit the fact that annihilation and creation operators are defined up to a global phase, we arrive at

$$\kappa = \eta$$

► Finally, rescaling the field to reabsorb the arbitrary overall constant we arrive at ... (*continued on the next page*).

Added explanation (not in the original notes)

Why the relative sign inside the bracket is a minus. The field contains a with weight κ and $a^\dagger$ with weight η ; hence $V^\dagger$ contains a with weight η^* and $a^\dagger$ with weight κ^* . In $[V^\mu(x), V^\nu(y)^\dagger]$ the surviving contractions are $\kappa\kappa^*[a, a^\dagger]$ (giving $|\kappa|^2 e^{-ip(x-y)}$) and $\eta^*\eta[a^\dagger, a]$ (giving $-|\eta|^2 e^{+ip(x-y)}$). The minus comes from the order reversal in the second commutator — it is *not* put in by hand. This is precisely why this bracket probes $|\kappa|^2 - |\eta|^2$ whereas the previous one probed $\kappa\eta$.

Physics of $|\kappa| = |\eta|$. This is the field-theoretic origin of **antiparticles**. Causality does not allow us to have a field that creates antiparticles more strongly than it destroys particles: outside the light cone the two amplitudes must cancel exactly. A particle and its antiparticle therefore have the same mass and the same spin, and are produced with the same strength.

Why the phase can be removed. A redefinition $a(\vec{p}, \lambda) \rightarrow e^{i\alpha} a(\vec{p}, \lambda)$ leaves all the (anti)commutators, the Hamiltonian and the number operator unchanged. Under it $\kappa \rightarrow \kappa e^{i\alpha}$ and $\eta \rightarrow \eta e^{-i\alpha}$, so the relative phase between κ and η can be set to zero: with $|\kappa| = |\eta|$ this gives $\kappa = \eta$. The remaining common factor is a normalization of the field itself, which is removed by rescaling $V^\mu \rightarrow V^\mu/\kappa$. After both steps, $\kappa = \eta = 1$.

One consequence to keep in mind. With $\kappa = \eta$ and $v^\mu = (\varepsilon^\mu)^*$, the field becomes *Hermitian*, $V^\mu(x)^\dagger = V^\mu(x)$. A Hermitian field describes a particle that *is its own antiparticle* (like the Z^0 , or the photon in the massless case treated in Part II). If instead one wants distinct particle and antiparticle (like the $W^\pm$), one starts from a *complex* field with independent a and $b^\dagger$; the argument above is unchanged and simply becomes the statement that the two species have equal mass and spin.

3 The real vector field and the free-field propagator [p. 20****]

$$V^\mu(x) = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\varepsilon^\mu(\vec{p}, \lambda) e^{-ip \cdot x} a(\vec{p}, \lambda) + \varepsilon^\mu(\vec{p}, \lambda)^* e^{ip \cdot x} a^\dagger(\vec{p}, \lambda) \right]$$

REAL VECTOR FIELD

$V^\mu(x)$ is an Hermitian operator.

Added explanation (not in the original notes)

Transcription note. In the handwriting the label next to the box reads “REAL SCALAR VECTOR FIELD”: the word “scalar” is written and then followed by “vector”, evidently a slip of the pen (the object is a four-vector field, not a scalar). The intended label is **real vector field** — “real” in the sense that the operator is Hermitian, exactly as for the real scalar field of Lecture 6.

What changed with respect to Lecture 9. Three things have happened at once: $\kappa = \eta = 1$ (previous page); the coefficient functions of the antiparticle piece are $v^\mu = (\varepsilon^\mu)^*$, so the same polarization vectors serve both terms; and the spin label σ is renamed λ . The result is the standard textbook expansion of the Proca field.

The free field propagator

We compute the free field propagator

$$G_F^{\mu\nu}(x - y) = \langle 0 | T \left[V^\mu(x) V^\nu(y) \right] | 0 \rangle$$

$\hookrightarrow$ Formally, the vacuum expectation value of the T -ordered product of two fields $V^\mu(x)$, $V^\nu(y)$.

$\hookrightarrow$ In principle, we could compute it by substituting the expression for $V^\mu(x)$ and $V^\nu(y)$ using the properties of creation and annihilation operators.

In practice, we use an alternative way that defines the propagator as the Green’s function of the equation of motion.

Added explanation (not in the original notes)

Why the Green’s-function route is easier. The direct route requires (i) splitting the T -product into the two time orderings, (ii) doing the two mode-sum integrals, (iii) recognizing the result as a contour integral in p^0 . All the spin structure has to be dragged along. The Green’s-function route instead solves an *algebraic* 4×4 problem in momentum space — the tensor structure comes out automatically — and the only physics input left is the choice of contour, i.e. the $+\mathrm{i}\varepsilon$ prescription, which encodes the time ordering. The two definitions agree precisely because T -ordering is what makes the vacuum two-point function satisfy the inhomogeneous (rather than homogeneous) equation, the $\delta^{(4)}(x - y)$ on the right-hand side coming from the time derivatives hitting the step functions $\theta(x^0 - y^0)$.

4 Equation of motion for a massive spin-1 field [p. 21****]

We derive the E.o.M. We know that

$$\begin{cases} (\square + M^2) V^\mu(x) = 0, \\ \partial_\mu V^\mu(x) = 0. \end{cases}$$

We elaborate the KG equation as follows:

$$\begin{aligned} \square V^\mu + M^2 V^\mu &= \partial_\nu \partial^\nu V^\mu + M^2 V^\mu = \partial_\nu (\partial^\nu V^\mu) - \underbrace{\partial^\mu (\partial_\nu V^\nu)}_{\substack{\text{we have just added} \\ \partial_\nu V^\nu = 0}} + M^2 V^\mu = 0 \\ \underbrace{\partial_\nu (\partial^\nu V^\mu - \partial^\mu V^\nu)}_{\equiv F^{\nu\mu}} + M^2 V^\mu &= 0 \end{aligned} \quad (\star)$$

$\rightsquigarrow$

$$\begin{aligned} \partial_\nu F^{\nu\mu} + M^2 V^\mu &= 0 \\ F^{\nu\mu} &= \partial^\nu V^\mu - \partial^\mu V^\nu \end{aligned}$$

E.o.M. for a
massive
spin-1
field

Obs. The equation $\partial_\nu F^{\nu\mu} + M^2 V^\mu = 0$ *automatically* contains the constraint $\partial_\mu V^\mu = 0$ since, by taking the four-divergence, we get

$$\underbrace{\partial_\mu \partial_\nu F^{\nu\mu}}_{=0} + M^2 (\partial_\mu V^\mu) = 0 \quad \rightsquigarrow \quad \partial_\mu V^\mu = 0.$$

To compute the propagator, we go back to eq. $(\star)$, and write it in the form

$$\begin{aligned} \partial_\nu \partial^\nu V_\rho g^{\rho\mu} - \partial^\mu \partial^\rho V_\rho + M^2 g^{\mu\rho} V_\rho &= 0 \\ \left[g^{\mu\rho} (\partial_\nu \partial^\nu + M^2) - \partial^\mu \partial^\rho \right] V_\rho &= 0 \end{aligned}$$

Added explanation (not in the original notes)

Reading the trick. The Klein–Gordon equation plus the constraint are *two* equations; the Proca equation $\partial_\nu F^{\nu\mu} + M^2 V^\mu = 0$ is a single equation equivalent to both. The manipulation is just “add zero in a clever form”: $\partial^\mu (\partial_\nu V^\nu) = 0$ is subtracted so that the two derivative terms assemble into the antisymmetric field strength.

Why $\partial_\mu \partial_\nu F^{\nu\mu} = 0$. $\partial_\mu \partial_\nu$ is symmetric in $\mu \leftrightarrow \nu$ while $F^{\nu\mu} = -F^{\mu\nu}$ is antisymmetric; the full contraction of a symmetric with an antisymmetric tensor vanishes identically.

Where the mass term breaks gauge invariance. $\partial_\nu F^{\nu\mu} = 0$ alone (the $M = 0$ case) is Maxwell’s equation, which is invariant under $V^\mu \rightarrow V^\mu + \partial^\mu \chi$ and does *not* imply $\partial_\mu V^\mu = 0$. The term $M^2 V^\mu$ destroys that invariance — and exactly for that reason it *forces* the constraint, removing one of the four components. This is the algebraic counterpart of the counting $4 - 1 = 3 = 2s + 1$ for $s = 1$. In Part II, when $M \rightarrow 0$, the constraint will no longer come for free and gauge invariance will reappear in its place.

The Lagrangian behind it (not in the notes). The Proca equation follows from

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} M^2 V_\mu V^\mu,$$

with the sign conventions of the course. Varying with respect to V_μ gives $\partial_\nu F^{\nu\mu} + M^2 V^\mu = 0$

directly.

5 The propagator as a Green's function [p. 22****]

$$\left[g^{\mu\rho}(\square + M^2) - \partial^\mu \partial^\rho \right] V_\rho(x) = 0$$

and we consider solutions to the equation

$$\left[g^{\mu\rho}(\square_x + M^2) - \partial_x^\mu \partial_x^\rho \right] G_{\rho\sigma}(x - y) = i \delta^\mu_\sigma \delta(x - y)$$

with, in Fourier space,

$$G_{\rho\sigma}(x - y) = \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x - y)} \tilde{G}_{\rho\sigma}(p), \quad \delta(x - y) = \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x - y)},$$

that in Fourier space reads

$$\left[g^{\mu\rho}(-p^2 + M^2) + p^\mu p^\rho \right] \tilde{G}_{\rho\sigma}(p) = i \delta^\mu_\sigma$$

We look for solutions in the form

$$\tilde{G}_{\rho\sigma}(p) = A g_{\rho\sigma} + B p_\rho p_\sigma$$

$$\left(g^{\mu\rho}(-p^2 + M^2) + p^\mu p^\rho \right) \left(A g_{\rho\sigma} + B p_\rho p_\sigma \right) \stackrel{!}{=} i \delta^\mu_\sigma$$

$$A \delta^\mu_\sigma (-p^2 + M^2) + B (-p^2 + M^2) p^\mu p_\sigma + A p^\mu p_\sigma + B p^2 p^\mu p_\sigma = i \delta^\mu_\sigma$$

$$A \delta^\mu_\sigma (-p^2 + M^2) + p^\mu p_\sigma (B M^2 + A) = i \delta^\mu_\sigma, \quad \text{from which}$$

Added explanation (not in the original notes)

The two cancellations, spelled out. Expanding the product term by term:

$$\underbrace{g^{\mu\rho} g_{\rho\sigma}}_{\delta^\mu_\sigma} A (-p^2 + M^2) + \underbrace{g^{\mu\rho} p_\rho p_\sigma}_{p^\mu p_\sigma} B (-p^2 + M^2) + \underbrace{p^\mu p^\rho g_{\rho\sigma}}_{p^\mu p_\sigma} A + \underbrace{p^\mu p^\rho p_\rho p_\sigma}_{p^2 p^\mu p_\sigma} B.$$

The $-p^2$ of the second term cancels against the $+p^2$ of the fourth (this is what the two strokes in the handwriting mean), leaving $B M^2 p^\mu p_\sigma$; adding the third term gives the coefficient $(B M^2 + A)$. Since δ^μ_σ and $p^\mu p_\sigma$ are independent tensor structures, their coefficients must match separately — that is the pair of equations on the next page.

Why the ansatz is complete. The only tensors available are $g_{\rho\sigma}$ and $p_\rho p_\sigma$ (the Levi-Civita symbol would need more vectors, and antisymmetric structures are excluded because the left-hand operator and δ^μ_σ are symmetric). So $A g_{\rho\sigma} + B p_\rho p_\sigma$ is the most general possibility, and finding A, B solves the problem exactly.

Note on notation. $\delta(x - y)$ here means the four-dimensional $\delta^{(4)}(x - y)$, as its Fourier representation $\int d^4 p / (2\pi)^4 e^{-ip(x-y)}$ makes clear.

6 The Feynman propagator and the Feynman rule [p. 23****]

$$\begin{cases} A(-p^2 + M^2) = i \\ M^2 B + A = 0 \end{cases} \longrightarrow \begin{cases} A = \frac{-i}{p^2 - M^2} \\ B = \frac{i}{M^2(p^2 - M^2)} \end{cases}$$

We thus get

$$\tilde{G}_{\rho\sigma}(p) = \frac{i}{p^2 - M^2} \left(-g_{\rho\sigma} + \frac{p_\rho p_\sigma}{M^2} \right)$$

We introduce the **Feynman prescription** for the poles.

FEYNMAN PROPAGATOR

$$G_F^{\mu\nu}(x-y) = \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x-y)} \frac{i}{p^2 - M^2 + i\epsilon} \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right)$$

FEYNMAN RULE

$$\mu \xrightarrow{p} \nu = \frac{i}{p^2 - M^2 + i\epsilon} \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right)$$

Added explanation (not in the original notes)

What the $+i\epsilon$ does. The denominator $p^2 - M^2$ vanishes on shell, at $p^0 = \pm E_p$, so the p^0 integral is ill-defined until the poles are moved off the real axis. Writing $p^2 - M^2 + i\epsilon = (p^0)^2 - E_p^2 + i\epsilon$, the poles sit at $p^0 \simeq \pm(E_p - i\epsilon')$: the positive-energy pole is pushed *below* the axis and the negative-energy one *above*. Closing the contour downwards for $x^0 > y^0$ then picks up only the positive-energy pole, and upwards for $x^0 < y^0$ only the other. That is exactly the T -ordering: particles propagate forward in time, antiparticles backward. Different prescriptions ($-i\epsilon$, principal value, $p^0 \rightarrow p^0 \pm i\epsilon$) give the anti-time-ordered, the principal-value, and the retarded/advanced Green's functions — all solve the same differential equation, and the boundary condition is the physics.

The $p^\mu p^\nu / M^2$ term is trouble at high energy. Unlike the scalar propagator, this one does *not* fall off as $p \rightarrow \infty$: the longitudinal piece tends to a constant. In a theory of massive vector bosons this leads to amplitudes growing with energy and to a violation of perturbative unitarity around the TeV scale — historically the argument that something else (the Higgs boson, or new strong dynamics) must appear. Note also that this term is singular as $M \rightarrow 0$: the massless limit of the massive propagator is *not* the photon propagator, another sign that Part II will need a genuinely new construction.

Sanity check on the tensor structure. $p_\mu \tilde{G}^{\mu\nu} \propto (-p^\nu + p^2 p^\nu / M^2)$, which on shell ($p^2 = M^2$) vanishes: the propagator is transverse on shell, exactly as the polarization vectors are ($p_\mu \epsilon^\mu = 0$). Off shell it is not, and that residual longitudinal part is the source of the behaviour just described.

Observation

From

$$\underbrace{\langle 0|T(V^\mu(x)V^\nu(y))|0\rangle}_{[V^\mu]^2} = \underbrace{\int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \frac{i}{p^2 - M^2 + i\epsilon} \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right)}_{[M]^{4-2} = [M]^2}$$

7 Mass dimension and a consistency check [p. 24****]

We extract the mass-dimension of the field in natural units:

$$[V^\mu(x)] = [M]$$

Added explanation (not in the original notes)

The counting. In natural units ($\hbar = c = 1$) lengths are inverse masses, so $[d^4p] = [M]^4$ while the propagator $\sim 1/(p^2 - M^2)$ carries $[M]^{-2}$; the exponential and the tensor structure are dimensionless. Hence the right-hand side has dimension $[M]^{4-2} = [M]^2$, and since the left-hand side is $[V^\mu]^2$ we get $[V^\mu] = [M]^1$.

Cross-check from the Lagrangian. The action is dimensionless, $[d^4x] = [M]^{-4}$, so $[\mathcal{L}] = [M]^4$. The mass term $\frac{1}{2}M^2 V_\mu V^\mu$ then requires $[M]^2[V]^2 = [M]^4$, i.e. $[V] = [M]$. Same answer. Note that this is the *same* dimension as a scalar field — and different from a Dirac field, which has $[\psi] = [M]^{3/2}$.

A consistency check (exercise)

For a free theory of massive spin 1 particle, the Hamiltonian takes the form

$$H_0 = \sum_{\lambda} \int \frac{d^3\vec{p}}{(2\pi)^3} E_p a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda)$$

We can check the consistency of the field theory description by computing

$$[H_0, V^\mu(x)] = -i\partial_t V^\mu(x)$$

Using for H_0 the expression above and for $V^\mu(x)$ the expansion in terms of creation and annihilation operators, the commutator must give $-i\partial_t V^\mu(x)$.

Added explanation (not in the original notes)

Where this identity comes from and how the exercise goes. In the Heisenberg picture $\partial_t \mathcal{O} = i[H, \mathcal{O}]$, so $[H_0, V^\mu] = -i\partial_t V^\mu$: the check is that the field built out of $a, a^\dagger$ really evolves with the Hamiltonian built out of the same operators — i.e. that the “particle” picture and the “field” picture agree.

The computation needs only

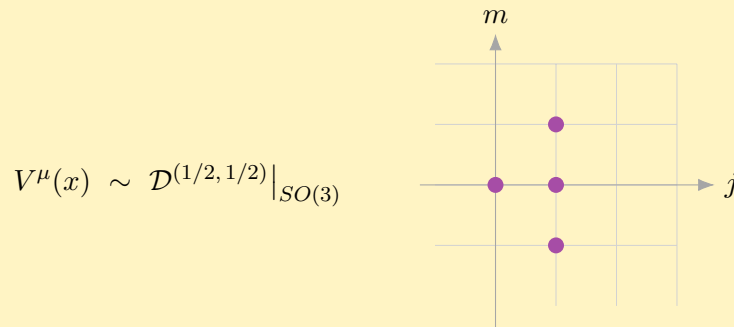
$$[a^\dagger(\vec{q}, \lambda') a(\vec{q}, \lambda'), a(\vec{p}, \lambda)] = -(2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta_{\lambda\lambda'} a(\vec{p}, \lambda),$$

and its Hermitian conjugate with $+$ for $a^\dagger$. Acting on the expansion of V^μ , each annihilation term picks up $-E_p$ and each creation term $+E_p$; since $\partial_t e^{\mp i p \cdot x} = \mp i E_p e^{\mp i p \cdot x}$, the two sides

match term by term. Note that the check is sensitive to the *commutators* established in Section 1: with anticommutators the signs would not work out, so this exercise is another (indirect) confirmation of Bose statistics.

8 Decomposition of a generic four-vector field [p. 24****-25****]

We saw that a four-vector field transforms under Lorentz in such a way that it contains under rotations a $j = 0$ component and a $j = 1$ component.



Added explanation (not in the original notes)

Reading the diagram. The horizontal axis is the total angular momentum j , the vertical axis its projection m . The single dot at $(j, m) = (0, 0)$ is the rotation-scalar V^0 ; the column of three dots at $j = 1, m = -1, 0, +1$ is the three-vector $\vec{V}$. Together they are the content of $\mathcal{D}^{(1/2, 1/2)}$ restricted to rotations, i.e. $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$: four states in total, matching the four components of V^μ .

More formally, this corresponds to the so-called Helmholtz decomposition

$$V^\mu(x) = V_T^\mu(x) + \partial^\mu \phi(x)$$

$$\partial_\mu V_T^\mu(x) = 0$$

A generic real four-vector field (4 independent components) can be decomposed into:

i) A four-vector field with vanishing four-divergence

$$V_T^\mu(x), \quad \partial_\mu V_T^\mu(x) = 0$$

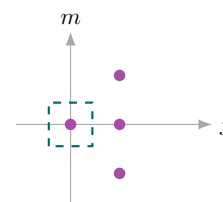
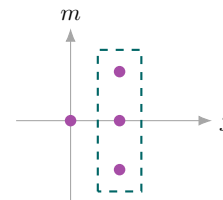
$\hookrightarrow$ **3 independent components**

This is the part we used to describe the massive spin-1 particles.

ii) The four-divergence of a scalar field

$$(\partial^\mu \phi)$$

$\hookrightarrow$ it carries the remaining **1 DOF**



Added explanation (not in the original notes)

Why this is called “Helmholtz”. The familiar three-dimensional Helmholtz theorem splits a vector field into a divergence-free (transverse) part and a curl-free (longitudinal) part, $\vec{V} = \vec{V}_T + \vec{\nabla}\phi$. The formula above is the four-dimensional, Lorentz-covariant version: $\partial^\mu\phi$ is the “pure gradient” piece and V_T^μ the “transverse” remainder. In momentum space the split is immediate: given $\tilde{V}^\mu(p)$, define $\tilde{\phi} = -i p_\nu \tilde{V}^\nu / p^2$ and $\tilde{V}_T^\mu = \tilde{V}^\mu - p^\mu p_\nu \tilde{V}^\nu / p^2$; the second is transverse by construction. (The $1/p^2$ is why the decomposition is non-local in position space.)

The counting, and why it matters. $4 = 3 + 1$. The transverse part carries the three states of a massive spin-1 particle; the gradient part carries one spin-0 state. So a generic four-vector field does not describe “a spin-1 particle” — it describes a spin-1 *plus* a spin-0 particle. The constraint $\partial_\mu V^\mu = 0$, which the Proca equation imposed for free, is precisely the statement “throw away the scalar”.

Preview of Part II. For $M = 0$ the counting must come out 2, not 3: a massless spin-1 particle has only two helicity states, $\lambda = \pm 1$. So one more component has to disappear, and the mechanism cannot be a constraint of the Proca type (there is no M^2 to produce it). The next part of the notes discovers where the missing reduction comes from — and the answer, foreshadowed by this decomposition, is that the $\partial^\mu\phi$ piece becomes *unphysical* rather than merely absent: it can be added to A^μ without changing anything. That is gauge invariance.

Part II — The particle–field connection: the massless case

9 The particle–field connection: the massless case [p. 1****]

Consider the equations derived in the massive case

$$\begin{cases} \mathcal{D}(\Lambda)^A_B u^B(m, \vec{p}, \sigma) = \sum_{\sigma'} \left[D_s(W(\Lambda, p)) \right]_{\sigma' \sigma} u^A(m, \vec{p}_\Lambda, \sigma') \\ \mathcal{D}(\Lambda)^A_B v^B(m, \vec{p}, \sigma) = \sum_{\sigma'} \left[D_s(W(\Lambda, p)) \right]_{\sigma' \sigma}^* v^A(m, \vec{p}_\Lambda, \sigma') \end{cases}$$

↓ In the massless case, these equations take the form

$$\begin{aligned} \mathcal{D}(\Lambda)^A_B u^B(m, \vec{p}, \lambda) &= e^{-i\theta(\Lambda, p)\lambda} u^A(m, \vec{p}_\Lambda, \lambda) \\ \mathcal{D}(\Lambda)^A_B v^B(m, \vec{p}, \lambda) &= e^{+i\theta(\Lambda, p)\lambda} v^A(m, \vec{p}_\Lambda, \lambda) \end{aligned}$$

We set a solution strategy based on **two steps**.

① We consider p as p_{REF} , and as Λ we take an element of the little group.

$$\begin{cases} p \rightarrow p_{\text{REF}} \\ \Lambda \rightarrow \Lambda_W \in SE(2) \end{cases} \implies p_\Lambda = \Lambda p \rightarrow \Lambda_W p_{\text{REF}} = p_{\text{REF}} = p_\Lambda$$

As a consequence of this choice, we have ...

Added explanation (not in the original notes)

What is different from the massive case, in one line. For $M > 0$ the little group of $p_{\text{REF}} = (M, \vec{0})$ is $SO(3)$, whose irreducible representations D_s are $(2s + 1)$ -dimensional *matrices* mixing the σ labels. For $M = 0$ the little group of $p_{\text{REF}} = \kappa \hat{z}$ is $ISO(2) \cong SE(2)$, the Euclidean group of the plane (Lectures 1–2!). Its physically acceptable unitary irreps — those with the continuous “translation” labels set to zero — are *one-dimensional*, labelled by the helicity λ , and act simply as the phase $e^{-i\theta\lambda}$ where θ is the angle of the $SE(2)$ rotation part. That is why the sum over σ' has collapsed into a single phase, and why the spin label σ has been replaced by λ .

Notation. The label m in $u^A(m, \vec{p}, \lambda)$ is kept from the massive notation (it was the mass); in the massless case it is of course zero, and one may simply read $u^A(\vec{p}, \lambda)$. $\theta(\Lambda, p)$ is the *Wigner phase*: the rotation angle of the little-group element $W(\Lambda, p) = H(p_\Lambda)^{-1} \Lambda H(p)$. The complex conjugate phase in the v equation is inherited from the D_s^* of the massive case.

Why this strategy works. These equations must hold for *every* Λ and every p . Rather than solving them in general, one picks two clever special cases: step ① takes Λ inside the little group, which fixes the *form* of the coefficient functions at the reference momentum; step ② takes Λ to be the standard boost itself, which then *generates* the coefficient functions at every other momentum for free. This is exactly the two-step method used for the massive case in Lecture 9.

10 Step ① and step ② [p. 2****]

$$\begin{aligned}\mathcal{D}(\Lambda_W)^A_B u^B(m, \vec{p}_{\text{REF}}, \lambda) &= e^{-i\theta(\Lambda_W, p_{\text{REF}})\lambda} u^A(m, \vec{p}_{\text{REF}}, \lambda) \\ \mathcal{D}(\Lambda_W)^A_B v^B(m, \vec{p}_{\text{REF}}, \lambda) &= e^{+i\theta(\Lambda_W, p_{\text{REF}})\lambda} v^A(m, \vec{p}_{\text{REF}}, \lambda)\end{aligned}$$

↪ We solve these equations and find the coefficient functions at reference momentum.

② We take again $p = p_{\text{REF}}$, but this time we take $\Lambda = H(\hat{p}, |\vec{p}|/\kappa)$ as Lorentz transformation. There are two consequences:

i) $p_\Lambda = \Lambda p$ becomes $p_\Lambda = \Lambda p = H(\hat{p}, |\vec{p}|/\kappa) p_{\text{REF}} = p$

$$p_\Lambda = p$$

ii) The Wigner phase is equal to zero. From the definition

$$\underbrace{H\left(\hat{p}_\Lambda, \frac{|\vec{p}_\Lambda|}{\kappa}\right)^{-1}}_{\substack{\text{with } p_\Lambda=p, \\ \text{this is } H(\hat{p}, |\vec{p}|/\kappa)^{-1}}} \Lambda \underbrace{H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)}_{\substack{\text{with } p=p_{\text{REF}}, \\ \text{this is the identity}}} = H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^{-1} H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right) = \mathbb{K}$$

⇒ consequently, the Wigner phase vanishes.

Added explanation (not in the original notes)

Untangling the notation of step ② — this page is the one place where the handwritten shorthand is genuinely confusing, so it is worth being pedantic.

The symbol p plays two roles. In the intertwining relation, the coefficient function on the left has argument p , and we set $p = p_{\text{REF}}$. The Lorentz transformation Λ is chosen to be the *standard boost that carries p_{REF} to some generic momentum p* , namely $\Lambda = H(\hat{p}, |\vec{p}|/\kappa)$, where now $\hat{p}$ and $|\vec{p}|$ refer to the *target* momentum. Then $p_\Lambda = \Lambda p_{\text{REF}} = p$ (the generic momentum) — this is what item i) says.

For the massless standard boost the convention is $p_{\text{REF}} = \kappa \hat{z} = (\kappa, 0, 0, \kappa)$ and

$$H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta \hat{z}), \quad e^\eta = \frac{|\vec{p}|}{\kappa},$$

i.e. first boost along $\hat{z}$ to change the energy from κ to $|\vec{p}|$, then rotate $\hat{z}$ onto $\hat{p}$. In particular, at the reference momentum itself $|\vec{p}_{\text{REF}}|/\kappa = 1$ and $\hat{p}_{\text{REF}} = \hat{z}$, so $H(\hat{z}, 1) = \mathbb{K}$ — that is the second underbrace.

Putting the three factors together: $W = H(\hat{p}, |\vec{p}|/\kappa)^{-1} \cdot H(\hat{p}, |\vec{p}|/\kappa) \cdot \mathbb{K} = \mathbb{K}$, hence $\theta = 0$. The intertwining relation then contains *no phase at all*, which is what makes step ② a pure generation formula.

The role of κ . In the massless case there is no mass to set the scale, so the reference momentum carries an arbitrary energy κ ; all physical results are independent of it. It plays exactly the role that M played in $p_{\text{REF}} = (M, \vec{0})$.

11 Generating the coefficient functions; the photon field [p. 3****]

The equations take the form

$$u^A(m, \vec{p}, \lambda) = \mathcal{D}(H)_B^A u^B(m, \vec{p}_{\text{REF}}, \lambda)$$

$$v^A(m, \vec{p}, \lambda) = \mathcal{D}(H)_B^A v^B(m, \vec{p}_{\text{REF}}, \lambda)$$

From $u^A(m, \vec{p}_{\text{REF}}, \lambda)$, $v^A(m, \vec{p}_{\text{REF}}, \lambda)$ we generate coefficient functions for generic momentum $\vec{p}$.

The case of the photon field

We consider the field

$$A^\mu(x) = \sum_{\lambda=\pm 1} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\varepsilon^\mu(\vec{p}, \lambda) e^{-ip \cdot x} a(\vec{p}, \lambda) + \varepsilon^\mu(\vec{p}, \lambda)^* e^{ip \cdot x} a^\dagger(\vec{p}, \lambda) \right]$$

Four remarks are attached to this expression in the notes:

- We **test the possibility** that massless spin-1 particles are described by a real four-vector field.
- We include **both helicities**, having in mind a parity-invariant theory like the QED.
- We introduce **polarization vectors** as coefficient functions.
- $E_p = |\vec{p}|$ in the massless case.

Added explanation (not in the original notes)

Why “test”. The word is deliberate. In the massive case the four-vector field *worked*: $\mathcal{D}^{(1/2,1/2)}$ contained $j = 1$, and the constraint $\partial_\mu V^\mu = 0$ trimmed four components to three. Here the target is two states, $\lambda = \pm 1$, and there is no obvious mechanism to remove two components covariantly. The rest of this lecture is the honest attempt — and the tension it uncovers is the whole point.

Why both helicities are put in by hand. For $M > 0$ the little group $SO(3)$ is connected and irreducibly links all $2s + 1$ values of σ : one irrep, one particle. For $M = 0$ each helicity λ is a *separate* one-dimensional irrep of $SE(2)$, so $\lambda = +1$ and $\lambda = -1$ are in principle two independent particles. What ties them together is *parity*, which flips $\lambda \rightarrow -\lambda$: a parity-invariant theory such as QED must contain both. (The weak interaction, which violates parity, is precisely where this is not automatic.)

Why $E_p = |\vec{p}|$. On the massless shell $p^2 = 0$ with $p^0 > 0$. The normalization factor $\sqrt{2E_p}$ is otherwise identical to the massive case.

12 Step 1a: Λ_W a rotation about $\hat{z}$ [p. 4****]

We directly work under the condition

$$v^\mu(\vec{p}, \lambda) = (\varepsilon^\mu(\vec{p}, \lambda))^*, \quad \text{which implies} \quad A^\mu(x)^\dagger = A^\mu(x).$$

We apply our strategy, starting from point ①. The equation is

$$\Lambda_W^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) e^{-i\theta(\Lambda_W, p_{\text{REF}})\lambda}$$

$$\Lambda_W \in SE(2)$$

Since $\Lambda_W \in SE(2)$, we have **two choices**:

1a) We consider as $\Lambda_W \in SE(2)$ a pure rotation around $\hat{z}$, $\Lambda_W \equiv \Lambda_R(\hat{z}, \phi)$.

$\hookrightarrow$ As a consequence, $\theta(\Lambda_R(\hat{z}, \phi), p_{\text{REF}}) = \phi$.

We thus have

$$\Lambda_R(\hat{z}, \phi)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = e^{-i\phi\lambda} \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$$

$$\hookrightarrow \Lambda_R(\hat{z}, \phi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi & 0 \\ 0 & \sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We then extract the following equations ...

Added explanation (not in the original notes)

Why $SE(2)$ splits into “rotations” and “translations”. Recall from Lecture 1 that the Euclidean group of the plane is generated by one rotation J^3 and two translations. Inside the Lorentz group, the little group of $p_{\text{REF}} = \kappa \hat{z}$ is generated by J^3 (rotations about the momentum direction — these leave p_{REF} untouched) together with the two combinations T_x, T_y written on [p. 8****]. So checking the little-group condition means checking it on these two kinds of elements separately: that is exactly the split 1a)/1b).

The rotation part is the easy half, and it is the one that carries the helicity information: $\theta(\Lambda_R(\hat{z}, \phi), p_{\text{REF}}) = \phi$ simply because the $SE(2)$ “rotation angle” of a pure rotation is the rotation angle itself. The phase $e^{-i\phi\lambda}$ is then the defining property of a helicity- λ state.

Why $A^{\mu\dagger} = A^\mu$ is imposed rather than derived. We are *testing* a real four-vector field, so Hermiticity is part of the hypothesis; taking $v^\mu = (\varepsilon^\mu)^*$ is what makes the two terms of the expansion mutual conjugates.

13 Transverse polarizations at reference momentum [p. 5****]

From the $\mu = 0$ and $\mu = 3$ rows:

$$\left. \begin{aligned} \varepsilon^0(\vec{p}_{\text{REF}}, \lambda) e^{-i\phi\lambda} &= \varepsilon^0(\vec{p}_{\text{REF}}, \lambda) \\ \varepsilon^3(\vec{p}_{\text{REF}}, \lambda) e^{-i\phi\lambda} &= \varepsilon^3(\vec{p}_{\text{REF}}, \lambda) \end{aligned} \right\} \longrightarrow$$

$$\varepsilon^0(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^3(\vec{p}_{\text{REF}}, \lambda) = 0, \quad \lambda = \pm 1$$

From the $\mu = 1, 2$ rows:

$$\begin{pmatrix} \varepsilon^1(\vec{p}_{\text{REF}}, \lambda) \\ \varepsilon^2(\vec{p}_{\text{REF}}, \lambda) \end{pmatrix} \underbrace{e^{-i\lambda\phi}}_{\substack{\cos(\lambda\phi) - i\sin(\lambda\phi) = \cos\phi - i\lambda\sin\phi \\ \text{since } \lambda = \pm 1}} = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} \varepsilon^1(\vec{p}_{\text{REF}}, \lambda) \\ \varepsilon^2(\vec{p}_{\text{REF}}, \lambda) \end{pmatrix}$$

$$\left\{ \begin{aligned} \varepsilon^1 \cancel{\cos\phi} - i\lambda \varepsilon^1 \sin\phi &= \cancel{\cos\phi} \varepsilon^1 - \sin\phi \varepsilon^2 \\ \varepsilon^2 (\cancel{\cos\phi} - i\lambda \sin\phi) &= \sin\phi \varepsilon^1 + \cancel{\cos\phi} \varepsilon^2 \end{aligned} \right. \longrightarrow \left\{ \begin{aligned} i\lambda \varepsilon^1(\vec{p}_{\text{REF}}, \lambda) &= \varepsilon^2(\vec{p}_{\text{REF}}, \lambda) \\ i\lambda \varepsilon^2(\vec{p}_{\text{REF}}, \lambda) &= -\varepsilon^1(\vec{p}_{\text{REF}}, \lambda) \end{aligned} \right.$$

$$i\lambda \varepsilon^1(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^2(\vec{p}_{\text{REF}}, \lambda), \quad \lambda = \pm 1$$

We fix the arbitrary normalization by setting

$$\varepsilon^1(\vec{p}_{\text{REF}}, \lambda) \equiv -\frac{\lambda}{\sqrt{2}} \quad \longrightarrow \quad \varepsilon^2(\vec{p}_{\text{REF}}, \lambda) = i\lambda \left(-\frac{\lambda}{\sqrt{2}} \right) = -\frac{i\lambda^2}{\sqrt{2}} = -\frac{i}{\sqrt{2}}$$

We get

TRANSVERSE POLARIZATIONS AT REF. MOMENTUM

$$\varepsilon^\mu(\kappa\hat{z}, \lambda = +1) = \frac{-1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ i \\ 0 \end{pmatrix} \quad \varepsilon^\mu(\kappa\hat{z}, \lambda = -1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -i \\ 0 \end{pmatrix}$$

Added explanation (not in the original notes)

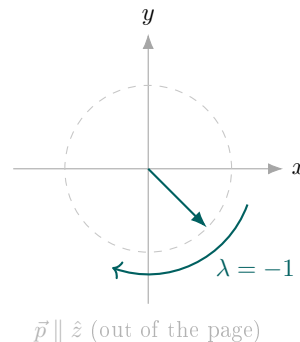
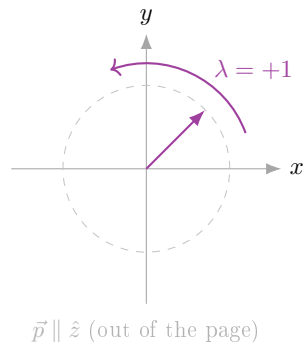
Step-by-step on the first pair of equations. $e^{-i\phi\lambda} \neq 1$ for generic ϕ , so $(e^{-i\phi\lambda} - 1)\varepsilon^0 = 0$ forces $\varepsilon^0 = 0$, and likewise $\varepsilon^3 = 0$. The reason these two rows decouple is visible in the matrix: $\Lambda_R(\hat{z}, \phi)$ acts as the identity on the 0 and 3 directions — a rotation about $\hat{z}$ touches only x and y . So the time component and the component *along the momentum* are killed immediately. *Only for $\lambda \neq 0$:* had $\lambda = 0$ been allowed, the phase would be 1 and nothing would follow. This is the first hint that $\lambda = \pm 1$ is special.

The identity $e^{-i\lambda\phi} = \cos\phi - i\lambda\sin\phi$ holds only because $\lambda = \pm 1$: then $\cos(\lambda\phi) = \cos\phi$ and $\sin(\lambda\phi) = \lambda\sin\phi$. It is what lets the whole ϕ dependence cancel, leaving a ϕ -independent condition.

Consistency of the two conditions. Multiplying $i\lambda\varepsilon^1 = \varepsilon^2$ by $i\lambda$ gives $-\lambda^2\varepsilon^1 = i\lambda\varepsilon^2$, i.e. $i\lambda\varepsilon^2 = -\varepsilon^1$: the second equation is not new. One complex condition on two complex numbers leaves one overall complex constant, fixed by the choice of normalization.

Check of the final vectors. For $\lambda = +1$: $\varepsilon^1 = -1/\sqrt{2}$ and $\varepsilon^2 = -i/\sqrt{2}$, i.e. $\varepsilon^\mu = -\frac{1}{\sqrt{2}}(0, 1, i, 0)$. For $\lambda = -1$: $\varepsilon^1 = +1/\sqrt{2}$ and $\varepsilon^2 = i(-1)(1/\sqrt{2}) = -i/\sqrt{2}$, i.e. $\varepsilon^\mu = \frac{1}{\sqrt{2}}(0, 1, -i, 0)$. They satisfy the standard normalization $\varepsilon(\lambda) \cdot \varepsilon(\lambda')^* = -\delta_{\lambda\lambda'}$ and $\varepsilon^\mu(\lambda)^* = -\varepsilon^\mu(-\lambda)$ — the sign convention (Condon–Shortley) is exactly what the choice $\varepsilon^1 \equiv -\lambda/\sqrt{2}$ implements.

What they mean physically: circular polarization. The spatial part is $\vec{\varepsilon}_\pm \propto \mp(\hat{x} \pm i\hat{y})/\sqrt{2}$. Taking the real part of $\vec{\varepsilon}e^{-i\omega t}$ gives a vector of constant length rotating in the xy -plane — the transverse plane perpendicular to the momentum — counter-clockwise for $\lambda = +1$ and clockwise for $\lambda = -1$. These are left- and right-circularly polarized light.



(Added figure: the real part of $\vec{\varepsilon}_{\pm} e^{-i\omega t}$ rotates in the plane transverse to $\vec{p}$; the two helicities rotate in opposite senses.)

14 Generic momentum: $\varepsilon^0(\vec{p}, \lambda) = 0$ [p. 6****]

We now consider the generic momentum

$$\begin{aligned}\varepsilon^\mu(\vec{p}, \lambda) &= H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right)^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) \\ &= \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\rho \underbrace{\Lambda_B(\eta \hat{z})^\rho{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)}_{\text{this is a pure spatial object with only } x \text{ and } y \text{ components,} \\ &\quad \text{it remains unaffected by a boost along } \hat{z}} \\ &= \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)\end{aligned}$$

We write

$$\varepsilon^\mu(\vec{p}, \lambda) = \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)$$

■ Consider the case with $\mu = 0$:

$$\varepsilon^0(\vec{p}, \lambda) = \underbrace{\Lambda_R(\hat{z} \rightarrow \hat{p})^0{}_\nu}_{\text{only the component with } \nu = 0 \text{ is non-vanishing}} \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \Lambda_R(\hat{z} \rightarrow \hat{p})^0{}_0 \underbrace{\varepsilon^0(\vec{p}_{\text{REF}}, \lambda)}_{=0} = 0$$

$$\varepsilon^0(\vec{p}, \lambda) = 0, \quad \lambda = \pm 1$$

At the level of the field operator, $A^0(x) = 0$.

Added explanation (not in the original notes)

Why the boost drops out. We just found $\varepsilon^0 = \varepsilon^3 = 0$ at the reference momentum, so $\varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)$ has only x and y components. A boost along $\hat{z}$ mixes only the 0 and 3 directions and acts as the identity on x and y ; hence $\Lambda_B(\eta \hat{z})$ leaves $\varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda)$ untouched. The entire boost dependence of the polarization vector evaporates — only the rotation $\Lambda_R(\hat{z} \rightarrow \hat{p})$ survives. (Contrast the massive case, where the longitudinal polarization $\varepsilon^\mu(\lambda = 0)$ *does* feel the boost and grows like E/M .)

Why $\Lambda_R(\hat{z} \rightarrow \hat{p})^0{}_\nu$ has only the $\nu = 0$ entry. A rotation is block diagonal: it acts as 1 on the time component and as a 3×3 orthogonal matrix on the spatial ones. So its first row is $(1, 0, 0, 0)$, and $\varepsilon^0(\vec{p}, \lambda) = \varepsilon^0(\vec{p}_{\text{REF}}, \lambda) = 0$.

Warning about $A^0 = 0$. This condition is manifestly *not* Lorentz covariant: a boost would generate a time component. That is the first concrete symptom of the trouble with describing a massless spin-1 particle by a four-vector field, and it is worth keeping in view while reading the next page.

15 Transversality and the counting of degrees of freedom [p. 7****]

We notice that, if we take $\vec{p}_{\text{REF}} = \kappa \hat{z}$, we have

$$\vec{p}_{\text{REF}} \cdot \vec{\varepsilon}(\vec{p}_{\text{REF}}, \lambda) = 0$$

$\hookrightarrow$ Since both $\vec{p}_{\text{REF}}$ and $\vec{\varepsilon}(\vec{p}_{\text{REF}}, \lambda)$ rotate to $\vec{p}$ and $\vec{\varepsilon}(\vec{p}, \lambda)$ with the same rotation, the scalar product is invariant, and we get

$$\vec{p} \cdot \vec{\varepsilon}(\vec{p}, \lambda) = 0, \quad \lambda = \pm 1$$

At the level of field operators, we find

$$A^0(x) = 0; \quad \vec{\nabla} \cdot \vec{A}(x) = 0$$

We started from a field with four real components $A^\mu(x)$.

↓ We found two constraints $A^0(x) = \vec{\nabla} \cdot \vec{A}(x) = 0$ that reduce the number of independent components down to **two**.

It seems appropriate to describe two physical DOF.

Added explanation (not in the original notes)

Why $\vec{p} \cdot \vec{\varepsilon} = 0$ becomes $\vec{\nabla} \cdot \vec{A} = 0$. In the mode expansion each term carries $e^{\mp i\vec{p} \cdot \vec{x}}$, so $\vec{\nabla}$ brings down $\mp i\vec{p}$; the condition $\vec{p} \cdot \vec{\varepsilon}(\vec{p}, \lambda) = 0$ therefore kills the divergence mode by mode. These two conditions together are known as the **Coulomb** (or **radiation**) **gauge**.

The counting works — and that is exactly what should make us suspicious. $4 - 2 = 2$, matching the two helicities. But note *how* the two conditions arose: not from an equation of motion (as $\partial_\mu V^\mu = 0$ did in the massive case, courtesy of $M^2 \neq 0$), but from the little-group condition itself. And neither $A^0 = 0$ nor $\vec{\nabla} \cdot \vec{A} = 0$ is Lorentz invariant: a boost does not preserve them. So we have a field whose defining conditions single out a frame, which is in tension with the covariant transformation law $A^\mu \rightarrow \Lambda^\mu_\nu A^\nu$ we started from.

Where this goes. The resolution — which the notes are building towards — is that no four-vector field can transform covariantly *and* carry only two helicity states. The best one can do is a field that transforms covariantly *up to a gauge transformation*,

$$A^\mu(x) \longrightarrow \Lambda^\mu_\nu A^\nu(\Lambda^{-1}x) + \partial^\mu \Omega(x),$$

and the physical content is then defined only modulo $\partial^\mu \Omega$. Compare the Helmholtz decomposition at the end of Part I: the $\partial^\mu \phi$ piece, which for $M > 0$ was simply set to zero, here becomes *pure gauge*. Gauge invariance is not an extra postulate — it is the price of combining masslessness, spin 1, and Lorentz covariance.

16 Step 1b: Λ_W a pure two-dimensional translation [p. 8****]

1b) We now consider as $\Lambda_W \in SE(2)$ a pure two-dimensional translation. We thus write

$$(\Lambda_W)^\mu_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = e^{-i\theta(\Lambda_W, p_{\text{REF}})\lambda} \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$$

with

$$\Lambda_W = \exp(-i\alpha T_x - i\beta T_y); \quad \alpha, \beta \in \mathbb{R},$$

$$\begin{cases} T_x = J_{4\text{VEC}}^1 + K_{4\text{VEC}}^2 \\ T_y = J_{4\text{VEC}}^2 - K_{4\text{VEC}}^1 \end{cases}$$

What is the Wigner phase?

The Wigner phase vanishes: $\theta(\Lambda_W, p_{\text{REF}}) = 0$

By definition, we have

$$\underbrace{H\left(\hat{p}_\Lambda, \frac{|\vec{p}_\Lambda|}{\kappa}\right)^{-1}}_{\substack{\text{with } p_\Lambda = p_{\text{REF}}, \\ \text{this is the identity}}} \underbrace{\Lambda}_{\substack{\Lambda \equiv \Lambda_W(\alpha, \beta) \text{ and, since } \Lambda_W \text{ is a little} \\ \text{group element, } \Lambda_W p_{\text{REF}} = p_{\text{REF}} \rightarrow p_\Lambda = p_{\text{REF}}}} H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right) \in SE(2)$$

with $p = p_{\text{REF}} \rightarrow H(p_{\text{REF}}, |\vec{p}_{\text{REF}}|/\kappa) = \mathbb{K}$

Added explanation (not in the original notes)

What T_x, T_y are. The subscript “4VEC” means: the generators taken in the four-vector (defining) representation of the Lorentz group, i.e. the explicit 4×4 matrices for rotations J^i and boosts K^i from Lecture 3. The combinations

$$T_x = J^1 + K^2, \quad T_y = J^2 - K^1$$

are precisely the two generators that annihilate $p_{\text{REF}} = \kappa(1, 0, 0, 1)$ — one checks $T_x p_{\text{REF}} = T_y p_{\text{REF}} = 0$ directly — and together with J^3 they close on the algebra of $SE(2)$:

$$[J^3, T_x] = iT_y, \quad [J^3, T_y] = -iT_x, \quad [T_x, T_y] = 0.$$

The vanishing of the last commutator is what makes T_x, T_y behave like the two translations of the Euclidean plane (Lecture 1) — hence the name.

Why the phase vanishes. The Wigner element is $W = H(p_\Lambda)^{-1} \Lambda H(p)$. Here $p = p_{\text{REF}}$ so the right factor is $\mathbb{K}$; $\Lambda = \Lambda_W$ is a little-group element so $p_\Lambda = p_{\text{REF}}$ and the left factor is $\mathbb{K}$ too. Hence $W = \Lambda_W$ itself, a *pure* translation — and the phase θ is by definition the coefficient of the J^3 (rotation) part of W , which here is absent. So $\theta = 0$. This is spelled out on the next page.

Why this step is the decisive one. Step 1a used only the rotation subgroup and gave a clean, sensible answer. Step 1b tests whether the four-vector really furnishes a representation of the *full* little group. The condition it produces, $\Lambda_W^\mu{}_\nu \varepsilon^\nu = \varepsilon^\mu$, says that the polarization vector must be strictly invariant under T_x, T_y — and that is what the explicit computation announced at the end of this scan will test.

17 The invariance condition and its physical meaning [p. 9****]

Consequently the combination

$$H\left(\hat{p}_\Lambda, \frac{|\vec{p}_\Lambda|}{\kappa}\right) \Lambda H\left(\hat{p}, \frac{|\vec{p}|}{\kappa}\right) \equiv \Lambda_W(\alpha, \beta).$$

Since in this case $\Lambda_W(\alpha, \beta)$ is a pure two-dimensional translation, the term proportional to J^3 is zero $\Rightarrow \theta(\Lambda_W, p_{\text{REF}}) = 0$.

We get

$$(\Lambda_W(\alpha, \beta))^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda)$$

$\hookrightarrow$ This equation makes a lot of physical sense. Remember that the physical states at reference momentum are defined in such a way that

$$T_x |\vec{p}_{\text{REF}}, \lambda\rangle = T_y |\vec{p}_{\text{REF}}, \lambda\rangle = 0,$$

which imply

$$\exp(-i\alpha T_x - i\beta T_y) |\vec{p}_{\text{REF}}, \lambda\rangle = |\vec{p}_{\text{REF}}, \lambda\rangle.$$

$\hookrightarrow$ This equation, therefore, reflects a property of the physical states.

Let us just check that everything is consistent as it seems to appear...

We compute explicitly the left-hand side of this equation:

* * *

(End of the Lecture 10 scan — the computation begins on the next sheet.)

Added explanation (not in the original notes)

Transcription note. In the displayed “combination” the first factor is written without the inverse; from the definition of the Wigner element (and from the identical formula on [p. 8****]) it should read $H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/\kappa)^{-1}$. Since both outer factors are the identity here, nothing in the conclusion changes.

Why $T_x |\vec{p}_{\text{REF}}, \lambda\rangle = 0$. This is the physical-state condition from the classification of the $SE(2)$ irreps (Lecture 2). The general unitary irreps of $SE(2)$ come in two families: those with a continuous parameter $\varrho > 0$ labelling the “radius” of the translation eigenvalues, and the degenerate ones with $\varrho = 0$. The first would give particles with a *continuous* internal degree of freedom, never observed; discarding them means setting the translation generators to zero on physical states, which leaves the one-dimensional helicity irreps. So the invariance of $|\vec{p}_{\text{REF}}, \lambda\rangle$ under $\exp(-i\alpha T_x - i\beta T_y)$ is exactly the statement that we kept only the sensible irreps.

What the check will find (a preview, so that the cliff-hanger is readable). Writing the 4×4 matrices explicitly, one finds for $p_{\text{REF}} = \kappa(1, 0, 0, 1)$

$$(\Lambda_W(\alpha, \beta))^\mu{}_\nu \varepsilon^\nu(\vec{p}_{\text{REF}}, \lambda) = \varepsilon^\mu(\vec{p}_{\text{REF}}, \lambda) + \frac{c(\alpha, \beta, \lambda)}{\kappa} p_{\text{REF}}^\mu,$$

i.e. the polarization vector comes back to itself *up to a term proportional to the momentum*, not exactly. The invariance demanded by the boxed equation therefore *fails* at the level of the four-vector — the four-vector is not a representation of the full little group.

The resolution is the one anticipated at [p. 7****]: the extra piece is harmless provided we declare ε^μ and $\varepsilon^\mu + c p^\mu$ to be *the same physical polarization*. In position space that is $A^\mu \rightarrow A^\mu + \partial^\mu \Omega$: gauge invariance. Everything then becomes consistent, at the price of accepting that A^μ is not an observable and that only gauge-invariant quantities (such as $F^{\mu\nu}$, or amplitudes contracted with conserved currents, $p_\mu \mathcal{M}^\mu = 0$) have physical meaning.

Summary of the logic of this lecture.

- *Massive*: causality $\rightarrow$ commutators (Bose) and $\kappa = \eta$ (antiparticles); the Proca equation delivers $\partial_\mu V^\mu = 0$ for free; $4 - 1 = 3$ states; propagator with a $p^\mu p^\nu / M^2$ term.
- *Massless*: little group $SE(2)$ instead of $SO(3)$; helicity phases instead of spin matrices; rotations about $\hat{z}$ give the two transverse polarizations and $4 - 2 = 2$ states; but the translations T_x, T_y cannot be satisfied by a genuine four-vector — which is where gauge invariance enters.